Sampling Experiment Record

Advanced Machine Learning · Week 5 · September 29, 2026

Use this worksheet with Lab A, Lab B, or Sampling Explorer. These are formative activities; no new graded assignment is introduced.

1. State the question before changing a slider

W1. Write the target, the function or filtering quantity, and one falsifiable prediction.

Field	Your record
Target / observed data	
Quantity to estimate	
Exact reference available in this benchmark	
One setting to change	
Settings and data held fixed	
Prediction and reason	
Seeds and repetition count	

Examples of experiment types: increase iid sample count; shift an IS proposal; vary MH step size; compare SIS with adaptive resampling at fixed observations. Changing observation noise also changes the statistical problem, so label it separately.

2. Keep cost and accuracy separate

W2. Fill at least two rows from controlled runs. Export CSV from the app and keep the file with the notebook.

Run	Method / changed setting	Estimate and exact reference	Error	Relevant diagnostic	Cost and seed
Baseline					
Comparison					
Repeated-seed summary					

For fixed-target experiments, record both the first moment and second moment. For MCMC, record post-warmup draws per chain, chain count, warmup, proposal scale, initialization, rank R-hat, ESS, and MCSE for the quantity of interest. For IS, record proposal settings and **weight** ESS.

Do not label direct iid oracle draws, target-density evaluations, conditional draws, and wall-clock seconds as identical budgets. State what was counted. Report repeated-seed behavior rather than selecting only the best run.

For filtering, use the same observations when comparing particle counts or thresholds. Record RMSE against the Kalman mean separately from RMSE against latent truth. Record resampling events and surviving initial ancestors as well as weight ESS.

3. Explain a failure, not just a score

W3. Complete this short evidence paragraph:

> I estimated _____ under target _____. I changed _____ while holding _____ fixed. I predicted _____ because _____. Across _____ seeds, I observed _____. The plot shows _____. The diagnostic _____ is informative about _____ but cannot establish _____. My next test would be _____.

A convincing explanation identifies the mechanism: missed support, unstable weights, slow movement, rejected-state mishandling, missed modes, weight degeneracy, or ancestor loss. A result that contradicts your prediction is useful when you explain it honestly.

Final app activity

Open Lab B, run the final Sampling Explorer cell, and inspect both tabs. Use the code guide to follow control → callback → algorithm → returned plot/CSV.

The generated Gradio URL is temporary. Keep the notebook link, your saved copy, the exact code tag, and the exported CSV. A screenshot alone omits settings and numerical data needed for reproduction.